

No exact D.E $\rightarrow$ Become into one using $\mu(x)$

"Integrating Process"

Example of what can be transformed

$$\underbrace{(3xy + y^2)}_{M(x,y)} + \underbrace{(x^2 + xy)}_{N(x,y)} \frac{dy}{dx} = 0$$

No is exact

$$\frac{dy}{dx} = 0$$

$$\frac{\partial M}{\partial y} = 3x + 2y ; \frac{\partial N}{\partial x} = 2x + y$$

They be not the same

I want this

$$\frac{\partial \hat{M}}{\partial y} = \frac{\partial \hat{N}}{\partial x}$$

②

$$\mu(x) \frac{\partial M}{\partial y} = N(x,y) \frac{dM}{dx} + \mu(x) \frac{\partial N}{\partial x}$$

Solve this problem to get $\mu(x)$ that makes this true

$$\frac{\mu(x) M(x,y) + \mu'(x) N(x,y)}{\hat{M}(x,y)} \frac{dy}{dx} = 0$$

$$\mu(x) M(x,y) + \mu'(x) N(x,y) \frac{dy}{dx} = 0 \cdot \mu(x)$$

③

$$\frac{\mu(x) \frac{\partial M}{\partial y}}{??} = N(x,y) \frac{dM}{dx} + \frac{\mu(x) \frac{\partial N}{\partial x}}{??}$$

0.5

$$\frac{dM}{dx} = \frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)}{N(x,y)} \mu(x)$$

$$\mu(x) = e^{\int \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) / N(x,y) dx}$$

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④ corroborate $\mu(x)$

$$\mu(x) M(x,y) + \mu'(x) N(x,y) \frac{dy}{dx} = 0 \cdot \mu(x)$$

$$\frac{\partial \hat{M}}{\partial y} = \frac{\partial \hat{N}}{\partial x}$$

Problem: $\frac{dy}{dx} + P(x)y = g(x)$

Solve this using this Method "M(x)" even when can be solved by linear stuff